

CO	COURSE OUTCOMES
CO1	Explain the fundamental principles of soft computing and analyze the structure and working of artificial neural networks.
CO2	Apply various neural network models, architectures and learning methods to solve basic computational problems.
CO3	Develop and analyze fuzzy logic-based systems using fuzzy sets, inference mechanisms and defuzzification techniques.
CO4	Apply supervised, unsupervised and reinforcement learning techniques to solve real-world data analysis problems.

Que No.	Questions	Marks	Target CO	Target KL
1.a	What happens if all weights in a neural network are initialized to zero?	1	CO1	L2
1.b	Why weights are randomly initialized in neural networks?	1		
1.c	Why are GPUs preferred for training neural networks?	1		
1.d	What happens if all inputs to a neuron are zero?	1		
1.e	Why neural networks are called "black box" models?	1		
2	Differentiate between a Biological Neuron and an Artificial Neuron. Explain at least five differences clearly.	5	CO1	L2
3	Explain the Perceptron model and its learning rule with necessary equations.	5	CO2	L2
4	Differentiate between Perceptron and ADALINE models with necessary equations.	5	CO2	L2
5	Explain the Backpropagation learning algorithm. Write the definite steps involved and include necessary equations. OR Consider a feedforward neural network The network uses a linear activation function $f(x) = x$. The learning rate $\eta = 0.1$ and bias = 0 (for all) $y = 1$ Perform one complete iteration of backpropagation and compute the updated values of all weights	10	CO2	L3

